

Muhammad Subhan Rauf

AI / Machine Learning Engineer

✉ subhan.rouf91@gmail.com | 📞 +92 3126904501 | [in linkedin.com/in/subhanrauf](https://www.linkedin.com/in/subhanrauf) | github.com/Mrup1

Summary

Computer Science graduate (FAST-NUCES) specializing in Large Language Models, Generative AI, and Machine Learning. Hands-on experience building production-oriented AI systems: RAG pipelines, LLM fine-tuning (LoRA / 4-bit quantization), multi-model LLM integration, and agentic workflows with LangChain and LangGraph. Comfortable owning the full backend pipeline from data acquisition to optimized inference, and shipping applications with FastAPI and Streamlit.

Technical Skills

- **Languages:** Python, C++, C, SQL
- **AI / ML:** LLM fine-tuning (LoRA, 4-bit / QLoRA quantization), Retrieval-Augmented Generation (RAG), prompt engineering, embeddings, chatbot integration, model deployment
- **Frameworks & Tools:** LangChain, LangGraph, LangSmith, FastAPI, Streamlit, Hugging Face Transformers, PyTorch, Scikit-learn, Pandas, NumPy
- **LLM APIs:** OpenAI (GPT), Anthropic (Claude), Google (Gemini), Hugging Face (Mistral-7B, MiniLM)
- **Platforms & Dev:** Git / GitHub, REST APIs, VS Code, Jupyter, Power BI, Draw.io

Experience

AI Backend Developer Intern

NUSyS Research Lab, FAST-NUCES

Backend & Team Manager

- Built backend components for an AI-driven therapeutic system based on Dual-Brain Psychology, delivered as FastAPI services.
- Integrated multiple LLMs (OpenAI GPT, Anthropic Claude, Google Gemini, Hugging Face) behind a unified API layer with model routing and fallback.
- Implemented a RAG framework and used LangChain / LangGraph for workflow orchestration; monitored token usage, cost, and latency via LangSmith.
- Served as Team Manager for the cohort, coordinating task delegation, progress tracking, and communication with research supervisors.

Projects

Clinic AI — WhatsApp Healthcare Receptionist (Agentic RAG)

- Built an AI receptionist bot for private clinics handling patient queries and appointment booking over WhatsApp, integrated with the Cliniko practice-management API.
- Designed the agent with LangGraph for multi-step orchestration, guardrails for safe responses, and observability for tracing conversations and tool calls.

EEG-based Brain-Computer Interface (BCI) for Number Recognition

[Mrup1/final-year-project-](#)

- Developed a BCI system using real-time EEG signal analysis to classify digits from brain signals.
- Engineered the full pipeline: data acquisition, signal filtering, feature extraction, and optimized ML inference.

YouTube Video Q&A — RAG Application

[Mrup1/langchain-portfolio](#)

- Built an interactive Streamlit app that extracts a YouTube transcript from a URL and answers questions about the video.
- Implemented the RAG pipeline with Hugging Face models — Mistral-7B-Instruct-v0.2 (chat) and all-MiniLM-L6-v2 (embeddings).

LLM Fine-Tuning for Entry-Test Preparation

[Mrup1/FineTunedModel](#)

- Fine-tuned Mistral-7B-Instruct-v0.2 using 4-bit quantization and LoRA on a custom dataset of past entry-test exams, improving domain answer accuracy.

Formula 1 Race Outcome Prediction

[Mrup1/Formula-1-Race-Outcome-Prediction](#)

- Built an ML model predicting F1 race outcomes from historical data using Pandas, NumPy, and Scikit-learn; handled cleaning, feature engineering, and evaluation.

Education

B.S. Computer Science (BSCS)

2022 – 2026

FAST National University of Computer and Emerging Sciences, Peshawar — CGPA: 3.2 / 4.0

F.Sc Pre-Engineering — Divisional Public School & College, Sahiwal

2020 – 2022

Matriculation (Science) — Divisional Public School & College, Sahiwal

2018 – 2020

Additional Training

- **Coursework:** LangChain & LangGraph specialization — building agentic and RAG-based LLM applications.
- **Soft Skills:** Problem solving, critical thinking, leadership, team management, adaptability.